Concept Note

1) Project Name

Predicting National Life Expectancy Through Healthcare Investment and Educational Attainment: A Data-Driven Analysis for SDG 3 and SDG 4

2) Overview

This project aims to examine the linear relationship between Sustainable Development Goal 3: Good Health and Well-being and Sustainable Development Goal 4: Quality Education, two of the United Nations Sustainable Development Goals. The primary focus of the study is to identify, through a data-driven approach, the impact of a country's education level—measured by average years of schooling—on the overall life expectancy of its population.

The analyses aim to demonstrate that investments in education contribute not only to individual development but also function as a key driver for improving national-level health indicators.

3) Background

Within the framework of global development goals, health and education are often considered as two independent domains. However, socio-economic data indicate that these two areas are closely interconnected and mutually reinforcing. It is widely recognized that differences in life expectancy between developed and developing countries are correlated not only with healthcare expenditures but also with levels of educational attainment and schooling rates.

This study establishes a foundation for exploring the determining influence of education on life expectancy by utilizing high-dimensional cross-country data.

4) Key Objectives / Business Objectives

a) Research Questions

1. How do healthcare spending and education affect national life expectancy?
2. What is the link between average schooling years and life expectancy across countries?
3. How do economic factors like GDP and income composition influence longevity?
4. Which dataset variables correlate most strongly with life expectancy?
5. Are there differences in life expectancy patterns between developed and developing nations?
6. Can healthcare and education data help predict life expectancy trends?

b) Key Steps

1. Collect and examine country-level health, education, and economic data.
2. Clean and preprocess the dataset, handling missing values and variable types.
3. Perform Exploratory Data Analysis (EDA) with descriptive statistics.
4. Visualize relationships using scatter plots, correlation matrices, and distributions.
5. Identify the most influential factors, focusing on healthcare and education.
6. Interpret results in the context of SDGs 3 and 4.

7. Suggest insights for future AI or machine learning models predicting life expectancy.

5) Methods and Workflow

a) Datasets

The WHO & UN Life Expectancy Dataset, an open-source dataset containing data from the World Health Organization, was used in this project. The dataset includes country-level indicators such as annual life expectancy, years of schooling, healthcare expenditures, and various economic variables.

b) Data Cleaning and Preprocessing

Columns that were expected to contain numerical values but appeared in incorrect formats (such as mixed date or text formats), including Schooling and Total Expenditure, were cleaned using custom functions and converted into appropriate numeric data types.

Missing values were identified and managed using suitable techniques during the analysis process, including methods such as dropna and numeric_only. Additionally, column names containing structural inconsistencies, such as trailing spaces, were normalized to ensure consistency throughout the dataset.

c) Exploratory Analysis and Modelling

Descriptive Statistics: The analysis revealed that the average life expectancy in developed countries is approximately 79.2 years, whereas in developing countries it is 67.1 years.

Correlation Analysis: A strong positive correlation of 0.75 was observed between years of schooling and life expectancy.

Visualizations:

Histograms: Global distributions of life expectancy and years of schooling were visualized.

Scatter Plots: A clear positive trend was observed, indicating that life expectancy tends to increase as the level of education rises.

Box Plots: Significant differences in both education and health indicators between developed and developing countries were highlighted.

Heatmap: Relationships among all variables were summarized using a correlation heatmap.

d) Deliverables

Cleaned Dataset: A fully processed dataset prepared for analysis after completing the data cleaning and preprocessing steps.

EDA Report: Comprehensive statistical analysis results supported by visualizations and graphical representations.

Feature Discovery: A proposed new index designed to measure the combined impact of investments in education and health.

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